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System and method for authenticating a user to provide a web service

Abstract

A method for authenticating a user for a web service includes receiving, at a first device, a request from a second device to authenticate the user for the web service and in response to the request, returning a code from the first device to the second device and generating, with the second device, a human-perceptible rendering including the code. The method further includes, capturing the human-perceptible rendering by a third device, providing an identifier of the user by the third device, identifying the code from the human-perceptible rendering, and granting access to the web service based on the code identified and the identifier of the user.

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Background/Summary

RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 16/714,729 entitled, "System and Method for Authenticating a User Via a Mobile Device to Provide a Web Service On A Different Computer System" filed by Suwat Phruksawan, Toby R. Kendall, Patrick Colin McGraw, Dominic Emile Caudell, Valery Zubovsky and Charles E. Gotlieb on Dec. 14, 2019, which is a continuation of U.S. patent application Ser. No. 15/400,935 entitled, "System and Method for Authenticating a User Via a Mobile Device to Provide a Web Service On A Different Computer System" filed by Suwat Phruksawan, Toby R. Kendall, Patrick Colin McGraw, Dominic Emile Caudell, Valery Zubovsky and Charles E. Gotlieb on Jan. 6, 2017, which claims the benefit of U.S. Provisional Patent Application Ser. No. 62/276,204, entitled, "Method and Apparatus for Providing a Web Service Using a Mobile Device" filed by Suwat Phruksawan, Toby R. Kendall, Patrick Colin McGraw, Dominic Emile Caudell, Valery Zubovsky and Charles E. Gotlieb on Jan. 7, 2016, and U.S. Provisional Patent Application Ser. No. 62/406,332, entitled, "Method and Apparatus for Providing a Web Service Using a Mobile Device Capturing Dual Images" filed by Jay Desai, Konstantinos P. Konstantidides, Suwat Phruksawan, Toby R. Kendall, Patrick Colin McGraw, Dominic Emile Caudell, Valery Zubovsky and Charles E. Gotlieb on Oct. 10, 2016, each having the same assignee as the present application and each is hereby incorporated by reference in its entirety.

FIELD OF THE INVENTION

(1) The present invention is related to computer software and hardware and more specifically to computer software and hardware for providing one or more web services.

BACKGROUND OF THE INVENTION

(2) Logging into a web site typically is performed by a requesting the web site's home page and then entering a user identifier and password. The user may then obtain web services from the web site. Other web services may be provided after the user authenticates himself or herself via other techniques. Such arrangements are suboptimal.

(3) For example, they can be subject to fraud when a key logging program or other such program is installed on the users' computer systems. The key logging program can pick up the website domain name the user requests, and can also pick up the user's password when typed by the user, and report such information to the malicious party. A malicious party can then go to the web site domain name and retype the password.

(4) The server systems that provide web services must be highly vigilant for such malicious parties, and must take extra precautions to detect such malicious parties, which makes the computing systems that run the servers less efficient.

SUMMARY OF INVENTION

(5) A system and method allows a user to register at a web site to which the user will request one or more web services using a smart device. The user downloads an app to the smart device and authenticates himself to the server via the app. The app has, or is issued, a secure, unique token to allow the app to identify to the server the particular copy of the app used by the user. The server associates the token with the account for which the user authenticated himself upon successful authentication of the user.

(6) In one embodiment, a device identifier is used in conjunction with the token. The device identifier is a unique identifier that is issued to the device, and the token is a long-lived token issued to the app upon successful authentication of the user. In such embodiment, from then on, the token consists of the device identifier sent to the server around the time of the authentication, and the token issued to the device. The device identifier may be checked by the server to ensure that it matches the device identifier for the account stored at the server, when the token for the account is

received, to assist in authenticating the long-lived token at the server.

(7) When the user requests a log in page from the server or otherwise requests a web service such as one or more web services requiring authentication, the server provides to the browser for display, user interface elements to allow the user to log in, for example, via a conventional user identifier and password, and displays an image that allows the user to log in via the app on their smart device. The image is a unique image that can be decoded or matched from among a larger set of images to identify a unique code for that image. Audio may encode the code and be used in place of, or in addition to, the image.

(8) The user is optionally instructed to authenticate himself to the app, for example, via biometrics, such as a fingerprint reader managed by the operating system of the device, and if the user successfully authenticates himself or herself, the user is instructed to perform the next step as described below. In the embodiment in which the user does not authenticate himself or herself, the user is instructed to perform the next step as described below.

(9) The user is instructed to use the device to take a photograph of the unique image displayed by the web site or allow the camera of the device to generate an image of image, and the image of the image is scanned. In the embodiment in which audio is used, the audio is captured and/or scanned, for example, using a microphone of the device or another device that is coupled to the device or the server. The device takes the photograph or otherwise captures the image or audio, and either the app decodes the code from the photograph or captured image or audio and uploads the code and token to the server, or the app uploads the photograph or audio file and token for the server to decode the photograph or match it to those issued and not expired. The server checks the code and, if not expired, may optionally perform further validation of the user (e.g. checking the IP address for one with which the user has successfully used to log in before, or checking the location of the device to ensure it is at or near a location at which the user has successfully logged in before or at or near a location corresponding to an address the user provided as part of the user's registration information), and if the code is not expired or invalid and the user is further validated (or if the code is not expired or invalid, if further validation is not performed), associates the user identifier associated with the token to the session corresponding to the code from the image, and invalidates the code to prevent further use. In one embodiment, invalidating the code may be performed by disassociating it from the session identifier.

(10) Computer code is included as part of the page provided in response to the original request for a log on page or for other web services. The computer code repeatedly queries the server to identify whether the session identifier is associated with a user identifier, or listens for a message that such association has taken place, or performs a combination of both of these. The session identifier is either provided to the server by the code, included with the request, such as the source IP address and port, or is retrieved by the server from the cookie. The frequency of repetition of the query may be reduced over time and may be stopped after a threshold amount of time, at which point the session identifier and code are also invalidated on the server.

(11) Once the server responds affirmatively, the computer code redirects the browser to request a different location on the server that attempts to validate the token, and if validated, the browser is redirected by the server to a web page to which the user would have been redirected had the user provided a user identifier and password to the user interface elements on the home page and pressed a submit button or otherwise authenticated to himself or herself, allowing conventional web services to be provided. Other methods, such as a single sign on protocol, including SAML may be used to allow the web site that provides the image and sign on user interface elements to be different from the web site that will provide the web service. The user is thus authenticated and logged in to the server, without providing any information to the home page user interface elements and without operating any user interface control on such home page, and/or may receive web services provided by the server that require a user authentication and are not provided without such authentication. If the token is invalid, the user is not given access to the web services.

(12) In one embodiment, the user enables such functionality of the app, without which, logging in via the app is not performed by the app as described above.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a block schematic diagram of a conventional computer system.
- (2) FIG. 2, consisting of FIG. 2A FIG. 2B and FIG. 2C, is a flowchart illustrating a method of providing one or more web services using an app on a device according to one embodiment of the present invention.
- (3) FIG. 3A is a block schematic diagram of a system for providing one or more web services using an app on a device according to one embodiment of the present invention.
- (4) FIG. 3B illustrates a server of FIG. 3A in more detail according to one embodiment of the present invention.
- (5) FIG. 3C illustrates a device of FIG. 3A in more detail according to one embodiment of the present invention.
- (6) FIG. 3D illustrates a computer of FIG. 3A in more detail according to one embodiment of the present invention.

DETAILED DESCRIPTION OF A PREFERRED EMBODIMENT

(7) The present invention may be implemented as computer software running on a conventional computer system, computer software embodied on a non-transitory storage media, or otherwise. Referring now to FIG. 1, a conventional computer system **150** for practicing the present invention is shown. Processor **160** retrieves and executes software instructions stored in storage **162** such as memory, which may be Random Access Memory (RAM) and may control other components to perform the present invention. Storage **162** may be used to store program instructions or data or both. Storage **164**, such as a computer disk drive or other nonvolatile storage, may provide storage of data or program instructions. In one embodiment, storage **164** provides longer term storage of instructions and data, with storage **162** providing storage for data or instructions that may only be required for a shorter time than that of storage **164**. All storage elements described herein may include conventional memory and/or disk storage and may include a conventional database. Other system elements may include a conventional processor. All elements of a system include any or all of at least one input, at least one output and at least one input/output.

(8) Input device **166** such as a computer keyboard or mouse or both allows user input to the system **150**. Output **168**, such as a display or printer, allows the system to provide information such as instructions, data or other information to the user of the system **150**. Storage input device **170** such as a conventional floppy disk drive or CD-ROM drive accepts via input **172** computer program products **174** such as a conventional floppy disk or CD-ROM or other nonvolatile storage media that may be used to transport computer instructions or data to the system **150**. Computer program product **174** has encoded thereon computer readable program code devices **176**, such as magnetic charges in the case of a floppy disk or optical encodings in the case of a CD-ROM which are encoded as program instructions, data or both to configure the computer system **150** to operate as described below.

(9) In one embodiment, each computer system **150** is a conventional SUN MICROSYSTEMS T SERIES SERVER running the ORACLE SOLARIS 11 or higher operating system commercially available from ORACLE CORPORATION of Redwood Shores, California, a PENTIUM-compatible personal computer system such as are available from DELL COMPUTER CORPORATION of Round Rock, Texas running a version of the WINDOWS operating system (such as XP, VISTA, 7 or 8) commercially available from MICROSOFT Corporation of Redmond Washington or a Macintosh computer system running the OS X operating system commercially

available from APPLE INCORPORATED of Cupertino, California and the FIREFOX browser commercially available from MOZILLA FOUNDATION of Mountain View, California or INTERNET EXPLORER browser commercially available from MICROSOFT above, although other systems may be used. Each computer system **150** may be a SAMSUNG GALAXY S5 commercially available from SAMSUNG ELECTRONICS GLOBAL of Seoul, South Korea running the ANDROID operating system commercially available from GOOGLE, INC. of Mountain View, California. Various computer systems may be employed, with the various computer systems communicating with one another via the Internet, a conventional cellular telephone network, an Ethernet network, or all of these.

(10) Referring now to FIG. 2, consisting of FIGS. 2A, 2B and 2C, a method of providing one or more web services to a user using an app on a device or the device itself is shown according to one embodiment of the present invention.

(11) The user registers a user identifier and password and other information described herein with a web site to allow the user to securely log in to that web site with those items **210**. Other authentication information may be used instead of a user identifier and password or in addition to them, such as personally identifiable or other information, such as first and last name, driver's license number, date of birth, or biometric information, which includes a voice print, a coded identification of the user's speech characteristics.

(12) Also as part of step **210**, in one embodiment, the user registers a fingerprint or other biometric characteristic of that user with the device **210**, which may be a conventional smart phone, tablet or other similar device. To do so, the user may use the camera, microphone and/or fingerprint reader on the device, which records biometric characteristics and assigns them as belonging to the user of the device, either on the device or on the server. In one embodiment, this function is performed by the operating system of the device, although in another embodiment, this function is performed by the app, which is described as being installed after this step, but may be installed before it.

(13) The user installs an application, referred to as an app, on the device **212** using conventional techniques, including an app store or a download to the device. The app has internally stored therein, a token that uniquely identifies the copy of the app installed, remains secure within it, and may only be transmitted encrypted, over a secure connection in one embodiment. The token may be retrieved from a server by the app after it is installed, or the app may be downloaded with the token pre-installed.

(14) The user provides **214** to the app the user identifier and password of that user that allows the user to log in to a website that may be reached via a browser on a computer system that is different from the device, though the user may also use a browser on the device to log in to the web site as well. The different computer system may be a desktop or laptop computer system. The app prompts the user for such log in information as part of step **214**. Other methods of identifying the user to the server via the app may be performed, such as by the user supplying information previously received by the server as registration information, such as any or all of their first and last name, date of birth, driver's license or social security number, or a unique code assigned to the user's account upon request, and provided to the user by the server via a website or text message sent to the device, or using other conventional techniques.

(15) Additionally, as part of step **214**, the app provides the user identifier and password, and the token, or other authentication information to a server, over a secure communication channel such as HTTPS, optionally after encrypting such information, and the server compares the user identifier and password with the one stored during the registration process of step **210** or issued to the user. Voice may be recorded, and encrypted and uploaded to the server over the secure connection instead of the user identifier and password and the server converts it to a voiceprint when received and compares the voiceprint with one received from the user and stored previously by the server.

(16) If the user identifier and password or voiceprints or other authentication information does not match those previously stored on the server as described above **216**, the method continues at step

214. In such case the prompt provided by the app may also inform the user that the authentication information did not match. If the user identifier and password or voiceprints or other authentication information do match **216**, the server associates the token received with the account corresponding to the user identifier and password or voiceprint **218**. Thus, the token then is a unique identifier that identifies the user of the account.

(17) In one embodiment, the device identifier is provided to the server in step **214** and the token is provided to the app by the server as part of step **218**, and each is unique across devices. The server stores the device identifier associated with the token as part of step **218**. Each time the token is provided from the app to the server as described herein, the app also provides the device identifier, which the server uses to authenticate the token by comparing the device identifier and the token to those stored to ensure they are both stored associated with the same account.

(18) In one embodiment, the user enables **220** logging into a web site using fingerprint or other biometric authentication or otherwise using the device as described herein, and in another embodiment, no such enabling function is needed, and thus, step **220** is skipped, as indicated by the dashed line in the figure.

(19) At any time after the steps above (either with or without step **220**) have been taken, the user may use a computer system such as one that is other than the device on which the app is installed to request to log in to a web site or otherwise request one or more web services, such as one requiring authentication as described herein **222**. This may be performed by the user using a browser on a conventional desktop or laptop computer to request a home page of the web site. In response to the request, the web site generates or selects a unique image corresponding to a unique code and a session identifier and stores them on the server associated with one another **230**. In one embodiment, the code from the image is used as the session identifier, although such dual use may be less secure, and so in another embodiment, the code is different from the session identifier. The unique image may be any image which can encode a unique identifier, such as an image that contains, or is, a QR code or an image with a conventional watermark, such as is commercially available from DIGIMARC of Beaverton Oregon. The image may be one that can be compared with other images that have an encoded file name (using the filename of the matched image as the code for the image, without revealing a code in the image itself) or may be decoded itself, for example, measuring the distance in the X and Y planes from the upper right corner of one or more features relative to the size of the image to identify the code, and using the locations of still other features that have known locations to correct for distortions from camera angles and the like. In one embodiment, a timestamp is retrieved from an operating system and stored associated with the code and session identifier on the server. Audio may be used to encode the code instead of an image or in addition to it, for example by embedding tones of a certain frequency at one or more particular points in time relative to a reference tone.

(20) The web site server then provides to the computer system from which the request was received any or all of a cookie with the session identifier or an alternate session identifier associated with the session identifier at the optionally encrypted, computer code that operates as described below, and a web page prompting the user for the user's user identifier and/or password and also provides the image generated or selected, which the browser at the user's computer system receives, displays the user interface elements and the image and operates the code, as described herein and below **232**. The audio may be provided and played by speakers of the computer system or another device associated with the computer system and in communication with it as part of step **232**. The method continues at step **234** and step **270** of FIG. 2B.

(21) In one embodiment, the session identifier is a unique code generated by the server to identify the session, though it may simply be the source IP address and the port from the request for the home page, and is stored on the server but not provided to the user's computer system.

(22) The user then uses the app to request the app to authenticate the user to the web site **234**, which may be performed by the user operating a user interface control on the app (e.g. a button) for

this purpose.

(23) In one embodiment, instead of the user requesting the app to authenticate the user to the web site, when the user requests the page from an IP address the user has previously used before to authenticate the user, the app is either automatically started by the server, or the user receives a notification that if clicked on, starts the app at the point of authentication of step **236**. In one embodiment, step **220** includes enabling this feature and then logging into the web site for the first time after enabling the feature from the IP address the user would like to bind to starting the app. After that time, any request for the home page of the web site made by anyone from that IP address will cause the server to signal the app and place itself into the same state in which it would be after step **234** without the user starting the app or the user directly signaling the app to inform it what the user wants the app to do.

(24) The app requests the user to authenticate himself via the fingerprint reader on the device **236** or by using a different technique such as authentication via iris or face recognition using the camera on the device and the app either requests the device to authenticate the user using biometric information (e.g. fingerprint) and report whether such authentication passed or failed, receives biometric information and uploads a representation of the biometric information to a server and requests the server to report whether the authentication passed or failed using registration information described above, or the app performs the authentication using conventional biometric techniques **238**. If biometric authentication fails **240**, the method continues at step **236** and otherwise **240**, the method continues at step **242**.

(25) It is noted that the fingerprint/biometric component of steps **236-240** may be skipped in one embodiment, so that just possession of the device or running the app, which may require authentication, is adequate to enable the web service to be provided, and thus, no authentication, or other forms of authentication, may be used in place of steps **236-240**. The 'no authentication' case is indicated by the dashed line in the Figure, bypassing steps **236-240**.

(26) Other forms of authentication of the user may be used instead of a fingerprint reader, such as submitting a user identifier and password that can be used to authenticate the user by the app or encrypted and uploaded to the server, whereupon the server decrypts the user identifier and password and compares it to the user's user identifier and password.

(27) At step **242**, the app requests the user to point a camera on the device towards the browser displaying the web page of web site on the display of the computer system and take a photo of, or otherwise capture, the image, using a user interface control of the app (e.g. a button) in one embodiment, or via automatic detection of the image in another embodiment. The method continues at step **250** of FIG. 2B.

(28) At step **250**, the app takes a photo of or otherwise captures the image in response to the user operating the user interface element, or the app detecting the image, and either the app scans and decodes the capture of the image and uploads the code from the image and the token to the server, or the app uploads some or all of the image and the token, and the server decodes the image **252**. The app may perform such detection, for example, because the user has pointed the camera of the device towards the screen, and the app, scanning the images provided by the camera, has detected the image by scanning for patterns associated with the image, such as a logo adjacent to, or part of, the image. Such information may be encrypted and uploaded via a secure connection. Decoding the image may be performed by reading information in the image or it may be performed by matching the image with one stored on the server that has a code associated with it, such part or all of the filename of that image. As noted, audio may be captured and decoded instead of the image and the code is then processed as described herein.

(29) Other ways of communicating a code between the website and an app may be employed. The web site may communicate such as via text message to send the code to the app and the app may receive the text message, or the user may read the text message and enter the code to the app. The web site can display the code in numeric form and the user could enter the code to the app. The app

would then send the code to the server as described herein. The phone number to which the message was sent would be supplied by the user as part of the registration information to the web site.

(30) The server checks the code and determines whether the code is not invalid (as described below) and it corresponds to a valid session and checks the token to determine if it is not invalid **254**. In one embodiment, a valid session is any session having a session identifier and in another embodiment, a valid session is one that has a session identifier and/or image code with a timestamp that is issued upon session creation that is not older than a threshold amount of time from the current time, retrieved from an operating system. As noted above the session identifier may be the image identifier or the code decoded from the image or another identifier. In one embodiment, the token is valid if not invalidated, which may occur if the user loses the device or otherwise indicates that the security of the account may have been compromised, for example by changing the password for the account, in which case all tokens associated with the account may be invalidated.

(31) If the code does not correspond to a valid session **256**, the server notifies the app and the app notifies the user, for example, to refresh the web page and repeat the process **264** and the method continues at step **222**. Refreshing the web page may cause the server to provide a different image or audio as described above.

(32) If the code does correspond to a valid session **256**, the server invalidates the code to prevent further use (the code otherwise being considered valid), and in one embodiment, the server optionally checks other information to indicate the authenticity of the user, such as whether the user has previously logged in from the same IP address as the was used to request the web page **258**. In such embodiment, the source IP address of the user computer system is stored associated with the session identifier as part of step **230** and retained for subsequent use. In one embodiment, the app sends last known location coordinates of the user with the token as part of step **252** and such information is used to identify whether the user has previously logged in from a location within a threshold distance of that location was detected previously via a login as described herein, or whether such location is within a threshold distance of an address (e.g. work address or home address) provided by the user during registration of step **210**. If the user does not have such an additional indicia of authenticity **260**, in one embodiment, the server invalidates the session identifier and/or code for the image and notifies the app, and the app notifies the user that the user must log in via a user identifier and password on the web page in a conventional manner. If the user is further authenticated **260**, the method continues at step **262**. In another embodiment, no such additional authentication is performed and steps **258** and **260** are skipped as shown by the dashed line in the Figure, and step **266** is not used.

(33) At step **262**, the server associates the user identifier associated with the token (and/or device identifier) with the session identifier, which may be the code corresponding to the image, or the source IP address and port for the computer system used to request the home page of the web site, or another identifier such as the alternate identifier stored in the cookie that is used to locate the session identifier on the server via its association therewith.

(34) Operation of the Computer Code Supplied with the Web Page and Server Response to Such Computer Code.

(35) At step **270**, the computer code, such as JAVASCRIPT code, running on the computer system, waits for a period of time, such as one second, then sends the session or other identifier (which, as described above, may be an assigned unique identifier, the code from the image, or the session identifier may be the source IP address and port that was provided as part of the initial request for the home page of the web site. In the last such case, the session identifier may be retrieved by the server instead of being sent by the computer code) and then requests status from the server. The initial period of time spent waiting may be longer than the period of time between subsequent checks, and such period between subsequent checks may become longer as the time since the computer code was deposited pass thresholds of time, as identified, for example, by the code

retrieving a system clock when deposited in step **232**, and then periodically checking to see if any threshold amount of time has passed. In one embodiment, when the period of time since the computer code was deposited exceeds the threshold of time used to invalidate the code as described above with respect to step **254**, the computer code will terminate and stop checking or performing the functions described herein.

(36) In response to the status request, the server checks **272** whether that session identifier/code has been associated with a user identifier as described herein and in step **262**. If such an association has not been made **274**, the method continues at step **270**, and otherwise **274**, the method continues at step **276**. In another embodiment, the computer code sends one such request or no requests, and waits for the server to send the indication. The server may check for the association periodically or may send the indication when the association is made.

(37) At step **276**, the server indicates to the computer code that the association has occurred, and may deposit a cookie corresponding to a unique session identifier it generates and associates with the user identifier at this stage, if not already performed, and the code redirects to a server page that validates the token **278** and the browser complies **280**.

(38) Other methods of allowing the user to be authenticated to one web site from another web site may be used (e.g. SAML), or the user may remain on the same web site, for example, without redirection. To validate the token, it is determined whether it has been associated with an account and/or checked to ensure it has not been revoked. In one embodiment, instead of validating the token or in addition to validating the token, an artifact, a type of token used for the transfer, is validated. The server attempts to validate **282** the artifact and/or token, and if the artifact and/or token is valid **284**, the server redirects **286** the user's browser to a web page that operates as if the user has logged in using a user identifier and password or otherwise provided the requested web service and the browser complies. The user is then given access to web services, some or all of which are performed for the user **288**. The validation of the token process may be skipped in one embodiment, so that the user is only redirected one time or no times, as indicated, for example, by the dashed line from step **276** in the Figure. The user is then given access to functions provided by the web site only to users who have logged in or otherwise authenticated themselves to a sufficient degree or to the originally requested web service, and any or all such web services are provided **284**. For example, the user is allowed to log in and make a trade or the user is allowed to make a trade that exceeds a dollar amount, even if the user was already logged in. In this last case, the authentication technique described herein can provide a higher level of authentication than an ordinary log in. If the token and/or artifact is invalid **284**, the user is redirected to a web page that displays an error message and the user is not logged into the web site and is not given access to such web services **290**.

(39) Any number of users may use the method and system of the present invention to receive or be denied web services or other functions.

(40) System.

(41) Referring now to FIG. **3A**, a system for providing one or more web services using an app on a device or the device itself is shown according to one embodiment of the present invention. The system operates as described herein.

(42) Each system element described herein labeled as storage includes conventional memory or disk storage and may include a conventional database. All other elements of the system described herein include conventional logic elements arranged to operate as described herein as a hardware processor, or they may include a conventional hardware process or system of multiple processors, along with software that causes such processor system to perform the active functions described herein. The lines shown coupling the various system elements may be any number of input/outputs, inputs or outputs.

(43) Server **300** includes a conventional web server or other type of server providing a service via the Internet. Device **350** includes a conventional smart device, such as a conventional smart phone

or tablet or any other computer system that can run an app. Computer **340** includes a conventional desktop or laptop computer system or any other computer system capable of running a conventional browser or another program providing similar functionality. A web service includes any service that is provided over the internet such as stock trading or any other service or any other service for which authentication may be preceded.

(44) FIG. **3B** illustrates server **300** in more detail according to one embodiment of the present invention. FIG. **3C** illustrates device **350** in more detail according to one embodiment of the present invention. FIG. **3D** illustrates computer **340** in more detail according to one embodiment of the present invention.

(45) Referring now to FIG. **3A-3D**, each such computer system **300, 340, 350** includes a conventional communication interface, such as a conventional TCP/IP communication interface **302, 352, 342** running suitable communications protocols such as TCP/IP, Ethernet or both, and each is coupled via input/output **301, 341, 351** to network **398** such as an Ethernet network, the networks of the Internet, or both and all communication between such computer systems **300, 340, 350** is made via the device's **300, 340, 350** respective input/output **301, 341, 351** of communication interface **302, 342, 352**. Some or all communications may be encrypted, for example, using HTTPS, or other conventional techniques. Each such computer system **300, 340, 350** also includes conventional a keyboard, mouse and monitor or display used to communicate with the user of such computer system. Each user may have any number of computers **340** and devices **350** and there may be any number of users.

(46) Registration manager **312** registers users by receiving or providing registration information, such as a user identifier and password, addresses of the user (work and/or home), cell phone number, or other conventional information provided by users when registering, such as the information described herein, and stores such information in server storage **308**. As noted, biometric information may be used to register a user and such information may be stored in place of, or in addition to, the user identifier and password.

(47) The user may also register his or her biometric information with the device **350**, such as by registering a fingerprint of the user to device biometric manager **368** on the device. Device biometric manager **368** may include a conventional fingerprint authentication facility that is supplied with device **350** by its manufacturer, or it may be a part of app **360** and use camera **374** or microphone **356** of device **350**. Although an app is described herein, the functionality of the app may be part of device **350** itself.

(48) The user installs app **360** onto device **350** from a conventional app store or server **300** onto device **350** using conventional techniques. App **360** may include a conventional token as part of the installation or may download one from the server upon installation as described above. The token may be downloaded upon authentication, in another embodiment.

(49) Device account authorization manager **364** provides a user interface to the user of device **350** to request the user to authenticate himself or herself, for example, by providing some of the registration information described above (e.g. user identifier and password, biometric information, driver's license number, etc.) or an identifier supplied by registration manager **312** for this purpose so that the app **360** may authenticate the user to server **300**. Such action may occur when the user starts the app for the first time, for example. The user supplies such information, which device account authorization manager **364** supplies to registration manager **312** along with the token. Registration manager **312** receives the information, attempts to locate the registration information it receives and matches it against the information stored in server storage **308**, as described above. If such information matches the information for an account stored in server storage **308**, registration manager **312** stores the token in server storage **308** associated with the account, and indicates to device account authorization manager **364** that the user has been authenticated.

(50) The user may at any time enable web services access via the device as described herein by indicating to user interface manager **366** that such access is desired. When it receives such

indication, user interface manager **366** stores into device storage **354** a flag that such access is enabled and also provides the flag and the token to registration manager **312**, which stores it into server storage associated with the account corresponding to the token. In another embodiment, such access is always enabled at such time as the user authenticates himself or herself the device **350**, and no such flag is employed.

(51) User Requests Web Service.

(52) The user uses browser **344** on computer system **340**, which includes any conventional computer system, including a desktop or laptop computer system, or a smart device to request a web service. To do so, the user may request a home page from a web site provided by server system **300** to request the web services provided after authentication of the user to the web site. Other web services may be provided, and authentication may be similarly performed, using facilities other than a home page. In one embodiment, if computer system **340** is a smart device, it is not the same device as is used to authenticate the user as described below.

(53) Web service provider **314** receives the request. Web service provider **314** may include a conventional web server. When it receives the request, web service provider **314** signals image/audio generator **316**, which generates and stores into server storage **308**, or selects, an image from those previously stored in server storage **308** as described above, and provides an identifier of the image or the image or identifier of the audio file or the audio, and the code to web service provider **314**, which provides the image or audio or retrieves it from server storage **308** and provides the image or audio to browser **344** for display or playback to the user on display **346** or speakers (not shown) as part of a web page that may also include user interface elements to allow the user to authenticate himself or herself via other techniques as described above. Web service provider **314** also provides computer code **378** that will run under browser **344**, which browser receives and stores into computer code storage **348**. Browser **344** displays the image on display **346** or plays back the audio and browser **344** runs the computer code **378** in computer code storage **348**, which may be conventional RAM or other forms of storage. The operation of computer code **378** is described below. Web service provider **314** stores into server storage **308** the code of the image or audio, the current date and time it retrieves from an operating system (not shown) and an identifier of the session (which may be a unique identifier it generates and stores on the computer **340** in the form of a cookie, or may be the IP address and port from which the request was received).

(54) In one embodiment, image/audio generator **316** generates a code, but does not need to generate the image or audio, and the code is provided to web services provider **314**. Web services provider **314** may display the code itself to the user to allow the user to enter the code, or send it to device **350** such as by texting it to a cell phone number of device **350** stored in server storage **308** by registration manager **312** with the other registration information as described above.

(55) The web page supplied by web service provider **314** may or may not provide instructions to the user to use app **360** on device **350** to take a photograph or scan the image displayed by display **346** or to use the microphone **356** on the device **350** or an external device (not shown) to record or capture the audio.

(56) In one embodiment, the user starts app **360** on device **350** using conventional techniques and indicates to user interface manager **366** that the user wishes to authenticate himself or herself to the web page or otherwise obtain the web service that will be provided when the user so authenticates himself or herself. In one embodiment, user interface manager **366** only allows the user to so indicate if the flag described above is enabled.

(57) In another embodiment, the web service provider **314** checks the IP address from which the request for the web service (e.g. the request for the home page) was received with an IP address stored with the account in server storage **308**. The IP address stored in server storage may be the IP address from which the user registered or otherwise authenticated himself or herself at a prior time, that is stored in server storage **308** by server registration manager **312** when the user registers or authenticates himself or herself via the web. The IP address used the last time to authenticate the

user for a prior instance of a web service, stored as described herein, may be used to identify the user only if there is only one user who last used that IP address in one embodiment. If a match is detected, web service provider **314** provides the account identifier to server notification manager **330** which uses a conventional app notification service to provide a notification to device **350**.

(58) In one embodiment, device notification manager **370** has previously registered with such app notification service, received an identifier to use with such service (or used an existing identifier may be used, such as a device identifier that uniquely identifies the device) and provided the identifier to server notification manager **330**, along with the token, and server notification manager **330** stores into server storage **308** the identifier associated with the account information to which the token corresponds.

(59) When it receives the account identifier, server notification manager **330** sends device notification service **358** the identifier corresponding to the account identifier and a service code that indicates the user should be treated as if the user had requested the app to authenticate himself or herself to a web page or obtain another web service. A device notification service **358** on device **350** receives the notification from the notification service and displays it to the user in a conventional fashion, such as using an icon in a status bar at the top of the display screen of device **350**. The user may use the user interface of the device notification service **358** on device **350** to select the notification, and the device notification service **358** starts app **360** if not already started and provides the service code to device notification manager **370**, which reads the service code and, based on the service code, signals user interface manager **366** with an indication that the user wishes to authenticate himself or herself to a web page or to obtain a web service.

(60) In one embodiment, web service provider **314** provides the code to server notification manager **330**, which provides the code to device notification manager **370** via the notification service described above, as part of, or all of, the service code, either using or not using the process described above. Device notification manager **370** may display the code to the user for reentry via a user interface element provided and received by device notification manager **370**, or may provide the code directly to image/audio processing manager **376**, which utilizes it as described below as if it had decoded it, without decoding it from an image.

(61) As noted, in one embodiment, sound can be used instead of an image, with the code embedded in the sound, and decoded from the sound by the device **350** as described herein or by an external device.

(62) When user interface manager **366** receives the indication from the user or the signal and indication from device notification manager **370**, in one embodiment, user interface manager **366** prompts the user to authenticate himself or herself to device **350** using the device biometric manager **368**. User interface manager **366** enables device biometric manager **368**, which authenticates the user using biometrics such as a fingerprint reader built into device **360** or any other conventional biometric reader, such as a microphone **356** for voice authentication or camera for iris or facial recognition, any or all of which may be considered to be part of device biometric manager **368**. Device biometric manager **368** authenticates the user, optionally sending a file of biometric information such as a voice file or photograph file to server biometric manager **318** and server biometric manager **318** authenticates the user using conventional biometric authentication techniques and indicates to device biometric manager **368** that the user is authenticated, or device biometric manager **368** authenticates the user using conventional biometric authentication techniques. Device biometric manager **368** indicates to user interface manager **366** whether the user is authenticated. User interface manager **366** may use other techniques to authenticate the user in other embodiments, or need not authenticate the user at all in still other embodiments.

(63) If the user is indicated to be not authenticated, user interface manager **366** so indicates to the user and requests the user to re-authenticate or cancel the operation. If the user is indicated to be authenticated, or is not to be authenticated, user interface manager **366** requests the user to hold camera **374** of device **350** up to display **346** or hold microphone **356** near speaker **345** of computer

system **340** and indicate when the user has done so via a user interface element that user interface manager **366** provides. When the user so indicates, user interface manager **366** signals camera **374** to turn on and/or take one or more images and store them into device storage **354**, and signals image/audio processing manager **376**. It isn't necessary to capture an image and store it as described, as it may be scanned, sent to image/audio processing manager **376** and decoded in real time using the output of camera **374**. It is noted that the authentication process may be performed using another technique as described above, by using device biometric manager **368** (which can receive a user identifier and password or other authenticating information as described above) and optionally server biometric manager **318**, or skipped entirely by user interface manager **366**, in which case device biometric manager **368** need not be used.

(64) When signaled, image/audio processing manager **376** either attempts to decode the code from the image or images from camera **374** or sound from microphone **356** until one or more are decoded or uploads the image or images and the token to server processing manager **320**, which attempts to decode the image or one of the images. Decoding may be performed by matching an image or audio to a stored set of images or audio whose filenames represent the code for the image or by reading information in the image or audio as described above. If server processing manager **320** decodes the image or audio, it so indicates to image/audio processing manager **376** and otherwise image/audio processing manager **376** decodes the image. If the image is decoded by image/audio processing manager **376**, image/audio processing manager **376** provides to server processing manager **320** the code and the token.

(65) Server processing manager **320** checks the token to determine if it is valid as described above (i.e. it has not been invalidated) and checks the code to determine whether it is valid as described above (e.g. using the system clock to test whether the code is older than the threshold amount of time allowed and has not been invalidated) and optionally checks the session to determine whether it is valid (e.g. the user has not abandoned it, as indicated by the lack of requests received from bind check manager **308** as described below, which bind manager **326** monitors and invalidates the session identifier in server storage **308** if an expected request is not received).

(66) In one embodiment, if the user logs into the website using registration manager **312** and changes their password for the account into which they logged in using registration manager **312**, registration manager **312** will identify the tokens associated with such account in server storage **308** and invalidate them. A system administrator may invalidate an individual token, for example, if the user reports the device lost or stolen.

(67) In one embodiment, image/audio processing manager **376** may optionally additionally provide to server processing manager **320** other information, such as the location or last known location of device **350** which it retrieves from an operating system (not shown) of device **350**, which may be identified using conventional GPS techniques. Additionally, in one embodiment, registration manager **312** maintains in server storage **308** the IP address of the user at the time the user initially registered, and the IP addresses of the last N times the user logged in. Other authentication manager **324** additionally stores the IP addresses of the last M times the user identifier was bound as described below and herein.

(68) In one embodiment, if server processing manager **320** determines any or all of the code, token or session are not all valid, it so indicates (along with identifiers of which were invalid) to image/audio processing manager **376**, which informs the user of the error and provides instructions to the user to inform the user what to do to correct the error.

(69) In one embodiment, if server processing manager **320** determines any or all of the code, token or session are all valid, it provides the user identifier corresponding to the token and the session identifier corresponding to the code to bind manager **326** in one embodiment, or provides any or all of the IP address of the user, the location information of the user and the account identifier corresponding to the token to other authentication manager **324** in another embodiment.

(70) Other authentication manager **324** receives such information and checks it against the

information stored for the user in server storage **308** as described above, for example, by any or all of: comparing the last known location information against previously registered location information as described above, comparing the IP address against previous IP addresses used to register, log in via username and password, or IP addresses used to log in via coded image as described above. In one embodiment, any or all of these are used to generate a confidence score, with the score being higher if the last known location of the device is near a registered address, higher if the IP address used to request the page containing the image is the same as was used to register the user, higher if the IP address used to request the page containing the image matches more of the set of IP addresses used to recently log in using a user identifier and password or the set of IP addresses used to log in or otherwise request one or more web services using the image, with one set of IP addresses optionally contributing to a greater score than the other set. Other authentication manager **324** compares the score against a threshold and if the score is above the threshold, signals bind manager **326** with the user identifier corresponding to the token and the session identifier associated with the image.

(71) If the score is below the threshold, other authentication manager **324** signals server processing manager **320** with the token and indicates that the authentication didn't pass, in which case server processing manager **320** notifies image/audio processing manager **376**, which notifies the user to authenticate himself or herself using the user's user identifier and password on the web site or other conventional technique and other identification manager **324** invalidates the code for the image in server storage **308**.

(72) When bind manager **326** receives the user identifier and session identifier, it binds them (i.e. associates them) in server storage **308** by associating them with one another, and stores the IP address used to request the image in a list of M most recently used IP addresses for the account corresponding to the user identifier, that it optionally maintains for each account in server storage **308**.

(73) Operation of Computer Code.

(74) Bind check manager **380** of computer code **378** waits as described above, and then sends the session identifier to bind manager **326**, which checks server storage **308** to see if the session identifier has been bound to (i.e. associated with) a user identifier as described herein. Bind manager **326** indicates to bind check manager **380** whether such binding has occurred and if no such binding has occurred, bind check manager **380** waits as described above and repeats the process. If such binding has occurred, bind check manager **380** requests browser **344** to redirect the user to validate manager **328**, which optionally validates the token and, if not validated, provides a web page in response informing the user of the error and requesting the user to authenticate himself or herself using another method. If validated, validate manager **328** redirects browser **344** to request the web service from web service provider **314**, such as a log in, which allows the user to perform functions on the web site (e.g. make a securities trade), for example that require the user be logged in or otherwise authenticated. In one embodiment, the web service is a securities trade above a threshold dollar amount, so that even if the user is already logged into the account, the authentication process using the device is performed as described herein before the trade is allowed to be initiated. Web service provider **314** checks the binding and then provides the requested web service provided a user identifier is bound to the session and in accordance with permissions granted to the user identifier via a lookup table stored in server storage **308**.

(75) Conventional single sign on techniques such as SAML may be used to allow the user to authenticate to the ultimate web site from the first web site.

(76) As noted above, it isn't necessary for bind check manager **380** to repeatedly query bind manager **326**, as it may query only one time, or not at all, bind manager **326** may send the indication that the binding has occurred when it performs such binding.

(77) It is also noted that one or both redirections may be omitted, and that the token need not be validated by validate manager **382** or an artifact may be validated in addition to or instead of the

token.

(78) In one embodiment, periodically, code validity manager 322 checks the codes issued for a session, but not invalidated, to determine whether they are expired based on the timestamp for the code and the current time it retrieves from an operating system. Code validity manager 322 invalidates such codes, for example, by marking the code invalid in server storage 308.

(79) It is noted that each user may have their own device and there may be any number of users who log in or otherwise request any number of web services as described above any number of times.

(80) In one embodiment, a charge is assessed for each code provided or each code that is used to authenticate a user, and such charges are paid. Quantity discounts may be applied based on the number of codes provided or used to authenticate the user, and may be applied based on the ratio of the number of codes used to codes provided.

(81) In this manner, the user is able to access web services on a session to which the user never authenticated himself or herself directly (e.g. via a user identifier and password, such authentication technique being made available via a user interface provided by web service provider on the same web page that is used to deliver the image or audio). Instead, the user authenticates himself or herself to a device, and then captures the image or audio containing the code, which is decoded and used to identify the session via a communication session between the device and the server that is separate from the one used to supply the web page. The token supplied by the device, separate from the computer system used to render the image or audio, is then used to identify the user as the user that should be bound to the session. The user does not authenticate himself or herself via input to the computer system or the web page it displays.

(82) Each system element may include a conventional hardware processor or hardware processor system or processor system or processor that is coupled to a hardware memory or hardware memory system or memory or memory system, each of these being conventional in nature. All system elements are structural: the only nonce word to be used herein is “means”. Each system element described herein may include computer software or firmware running on a conventional computer system. Each system element labeled “storage” may include conventional computer storage such as memory or disk and may include a conventional database. Each system element may contain one or more inputs, outputs and/or input/outputs to perform the functions described herein. Any system element may incorporate any of the features of the method and vice versa.

Certain Embodiments

(83) Described is a method of authenticating a user for a web service including: receiving at a first device a request from a second device to authenticate the user for the web service; in response to the request, returning a code from the first device to the second device, in a form that, when rendered by the second device is rendered by the second device in a human perceptible way, with the code encoded in such rendering; by a third device, different from the first device and the second device, capturing a human perceptible rendering performed by the second device of the encoded code and providing an identifier of the user after the third device has authenticated the user; identifying the code from the human-perceptible rendering captured; and granting access to the web service to be provided using the first device, responsive to the code identified and the identifier of the user.

(84) The method may include an optional feature whereby the human perceptible rendering captured is at least a part of an image.

(85) The method may include an optional feature whereby the human perceptible rendering captured is at least a part of audio.

(86) The method may include an optional feature whereby the web service is granted additionally responsive to at least one action performed by the second device.

(87) The method may include an optional feature whereby the access is granted in response to two or more queries made by computer code executing on the second device that is deposited onto the

second device by the first device in response to the request.

(88) The method may include an optional feature whereby the second device authenticates the user via a biometric authentication.

(89) Described is a system for authenticating a user for a web service including: a web service provider at a first device, the web service provider having an input coupled to a second device over a network, the web service provider input for receiving a request from a second device to authenticate the user for the web service, the web service provider for identifying the second device at an output responsive to the request; and an image/audio generator having an input coupled to the web service provider for receiving the signal, the image/audio generator for providing at an output, in response to the signal, a set of information containing an encoded code that can be rendered by the second device in a human perceptible way with the code encoded in such rendering; and whereby the web service provider input is additionally coupled to the image/audio generator output for receiving the set of information, the web service provider output is additionally coupled to the second device, and the web service provider is additionally for providing at the web service provider output, the set of information to the second device in response to the request; a device account authorization manager at a third device, separate from the first device and the second device, and having an input for receiving authentication information, the device account authorization manager for authenticating the user responsive to the authentication information, and for providing at an output a token identifying the user in response to the authenticating; one selected from the group including A and B: A. an image/audio processing manager, at the third device, having an input coupled to the device account authorization manager output for receiving the token and to a camera or microphone of the third device, the image/audio processor for capturing via the camera or microphone a human perceptible rendering of the set of information performed by the second device where the rendering has the code encoded therein, the image/audio processing manager for identifying the code from the human-perceptible rendering captured and providing the code and the token at an output; and a server processing manager having an input coupled to the image/audio processing manager for receiving the code and token for providing the code and token at an output; and B. an image/audio processing manager, at the third device, having an input coupled to the device account authorization manager output for receiving the token and to a camera or microphone of the third device, the image/audio processor for capturing via the camera or microphone a human perceptible rendering of the set of information performed by the second device where the rendering has the code encoded therein, the image/audio processing manager for providing at an output at least some of the human-perceptible rendering captured and the token; and a server processing manager having an input coupled to the image/audio processing manager for receiving the at least some of the human-perceptible rendering captured and the token, the server processing manager for identifying the code from the at least some of the human-perceptible rendering received and providing the code and token at an output; and whereby the web service provider input is additionally coupled to the server processing manager output for receiving the code and token and for providing the web service responsive to the code and the token.

(90) The system may include an optional feature whereby the human perceptible rendering captured is at least a part of an image.

(91) The system may include an optional feature whereby the human perceptible rendering captured is at least a part of audio.

(92) The system may additionally include a bind check manager on the second device for, performing at least one action via an input/output; and whereby the access is granted additionally responsive to the at least one action.

(93) The system may include an optional feature whereby: the bind check manager includes computer code deposited on the second device by the web service provider via the web service provider output in response to the request; and the at least one action includes two or more queries made by the bind check manager operating on the second device.

- (94) The system may include an optional feature whereby the authentication information comprises biometric information.
- (95) Described is a computer program product including a computer useable medium having computer readable program code embodied therein for authenticating a user for a web service the computer program product including computer readable program code devices configured to cause a computer system to: receive at a first device a request from a second device to authenticate the user for the web service; in response to the request, return a code from the first device to the second device, in a form that, when rendered by the second device is rendered by the second device in a human perceptible way, with the code encoded in such rendering; by a third device, different from the first device and the second device, capture a human perceptible rendering performed by the second device of the encoded code and providing an identifier of the user after the third device has authenticated the user; identify the code from the human-perceptible rendering captured; and grant access to the web service to be provided using the first device, responsive to the code identified and the identifier of the user.
- (96) The computer program product may include an optional feature whereby the human perceptible rendering captured is at least a part of an image.
- (97) The computer program product may include an optional feature whereby the human perceptible rendering captured is at least a part of audio.
- (98) The computer program product may include an optional feature whereby the web service is granted additionally responsive to at least one action performed by the second device.
- (99) The computer program product may include an optional feature whereby the access is granted in response to two or more queries made by computer code executing on the second device that is deposited onto the second device by the first device in response to the request.
- (100) The computer program product may include an optional feature whereby the second device authenticates the user via a biometric authentication.

Claims

1. A system for authenticating a user for a web service comprising: a web service provider comprising a hardware computer processor coupled to a computer memory at a first device, the web service provider having an input coupled to a second device over a network, the input of the web service provider configured to receive a request from the second device to authenticate the user for the web service, the web service provider configured to provide a signal at an output responsive to the request; and an image/audio generator comprising the hardware computer processor coupled to the computer memory having an input coupled to the output of the web service provider, the input of the image/audio generator configured to receive the signal, the image/audio generator for providing at an output, in response to the signal, a set of information including a human-perceptible rendering including a code; and wherein the input of the web service provider is additionally coupled to the output of the image/audio generator and is configured to receive the set of information, the output of the web service provider is additionally coupled to the second device, and the web service provider is additionally configured to provide, at the output of the web service provider, the set of information to the second device in response to the request; a device account authorization manager at a third device, separate from the first device and the second device, the device account authorization manager for providing at an output a token identifying the user; the system further comprising elements selected from a group consisting of A and B: A. an image/audio processing manager, at the third device, having an input coupled to the output of the device account authorization manager for receiving the token and to a camera or microphone of the third device, the image/audio processing manager for capturing, via the camera or the microphone, the human-perceptible rendering of the set of information performed by the second device, the image/audio processing manager further configured to identify the code from

the human-perceptible rendering captured by the image/audio processing manager and provide the code and the token at an output; and a server processing manager having an input coupled to the output of the image/audio processing manager for receiving the code and the token and an output for providing the code and the token; and B. an image/audio processing manager, at the third device, having an input coupled to the output of the device account authorization manager for receiving the token and to a camera or microphone of the third device, the image/audio processing manager for capturing, via the camera or the microphone, the human-perceptible rendering of the set of information performed by the second device, the image/audio processing manager further configured to provide at an output at least some of the human-perceptible rendering captured by the image/audio processing manager and the token; and a server processing manager having an input coupled to the output of the image/audio processing manager for receiving the at least some of the human-perceptible rendering captured by the image/audio processing manager and the token, the server processing manager configured to identify the code from the at least some of the human-perceptible rendering received and provide the code and the token at an output; and wherein the input of the web service provider is additionally coupled to the output of the server processing manager for receiving the code and token and the web service provider is additionally configured to authorize, via the output of the web service provider, the providing of the web service responsive to the code and the token.

2. The system of claim 1, wherein the human-perceptible rendering captured by the image/audio processing manager is at least a part of an image.

3. The system of claim 1, wherein the human-perceptible rendering captured by the image/audio processing manager is at least a part of audio.

4. The system of claim 1: additionally comprising a bind check manager on the second device for performing at least one action via an input/output; and wherein the web service is provided additionally responsive to the at least one action.

5. The system of claim 4, wherein: the bind check manager comprises computer code deposited on the second device by the web service provider via the output of the web service provider in response to the request; and the at least one action comprises a plurality of queries made by the bind check manager operating on the second device.

6. The system of claim 1, wherein the elements comprise A.

7. The system of claim 1, wherein the elements comprise B.
